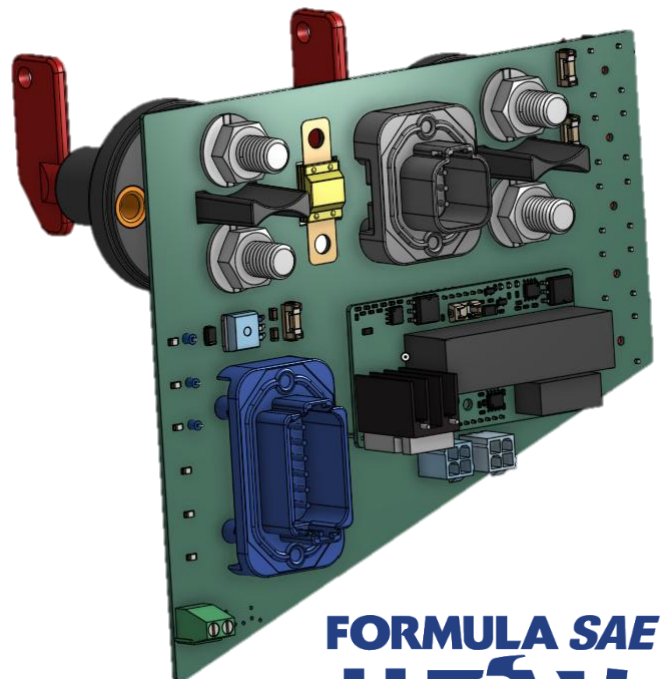
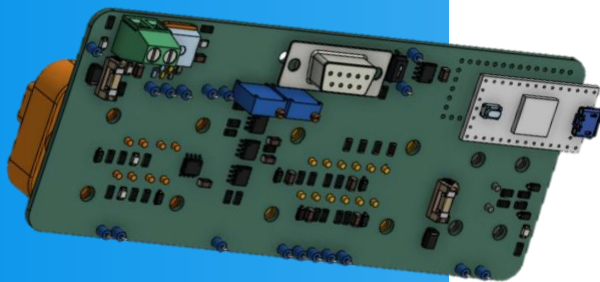
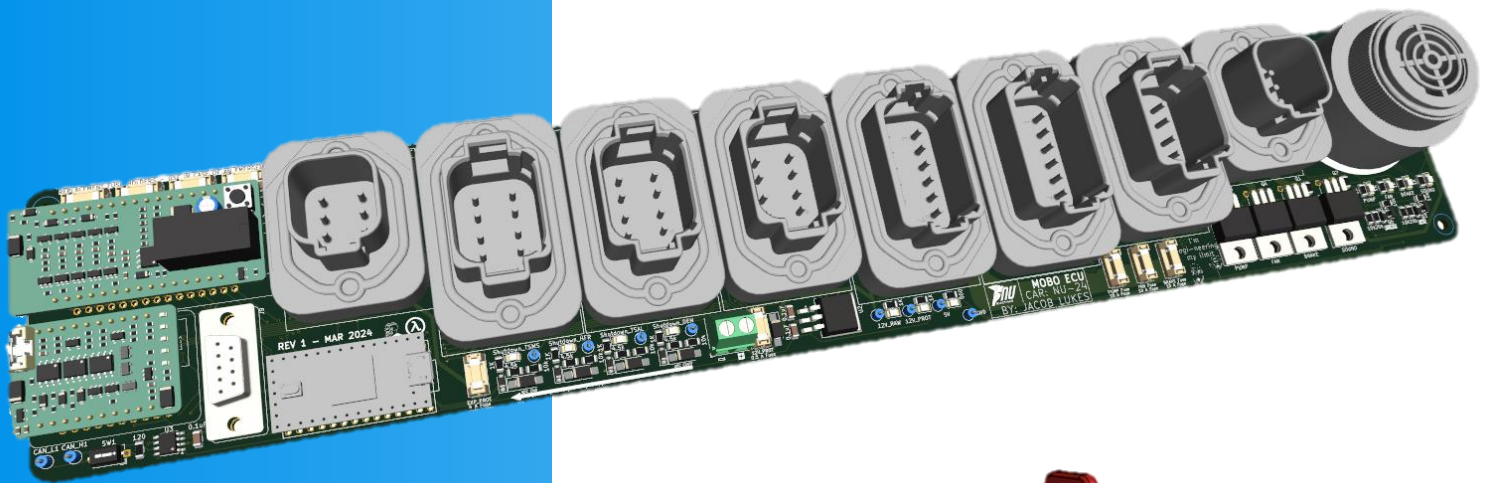




# How To Import PCBs Into OnShape

ZERO-TO-HERO



Jacob Lukes  
2025





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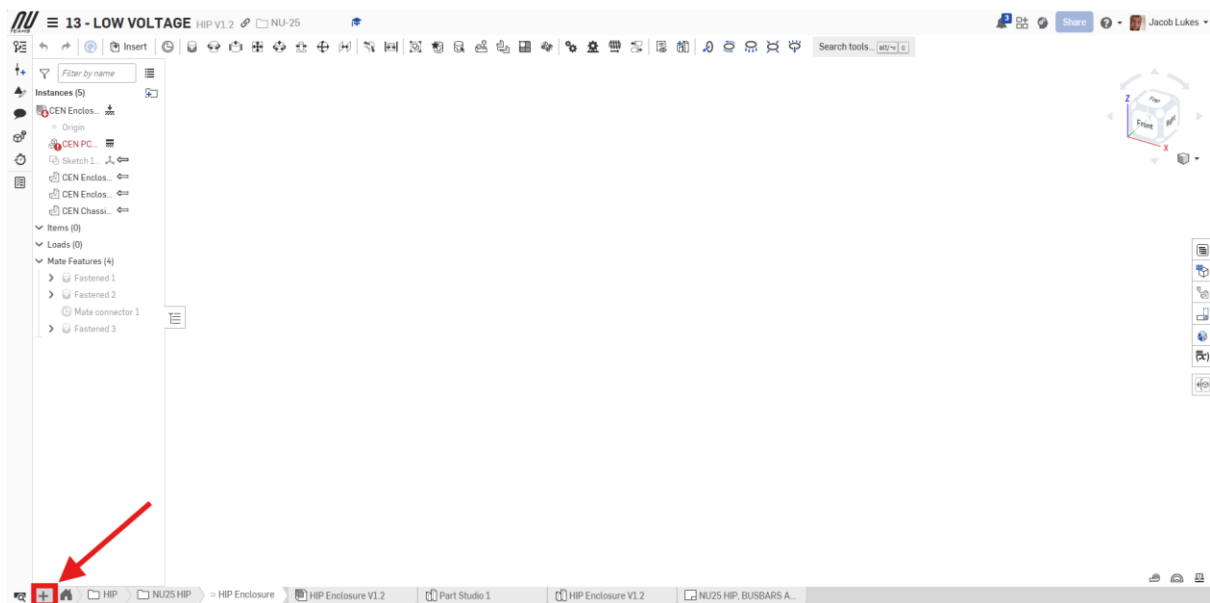
## Sanity Checks

- Firstly, ensure **ALL** step files for footprints are present and correctly positioned
  - This includes:
    - All dimensions for components are correct, mainly large components that protrude from the board. It is the PCB designer's responsibility to know this
    - Checking that mating connectors and attached wires are accounted for
      - eg: Micro/Mini Fit MOLEX connectors that attach to the board in any way
      - This can be as simple as editing the original STEP file for the connector, and extruding it
  - Getting this part right saves **LOTS** of time down the line for the individual making the enclosure and the team as a whole
  - Refer to PCB Design Zero-To-Hero on how to make and attach STEP files to footprints

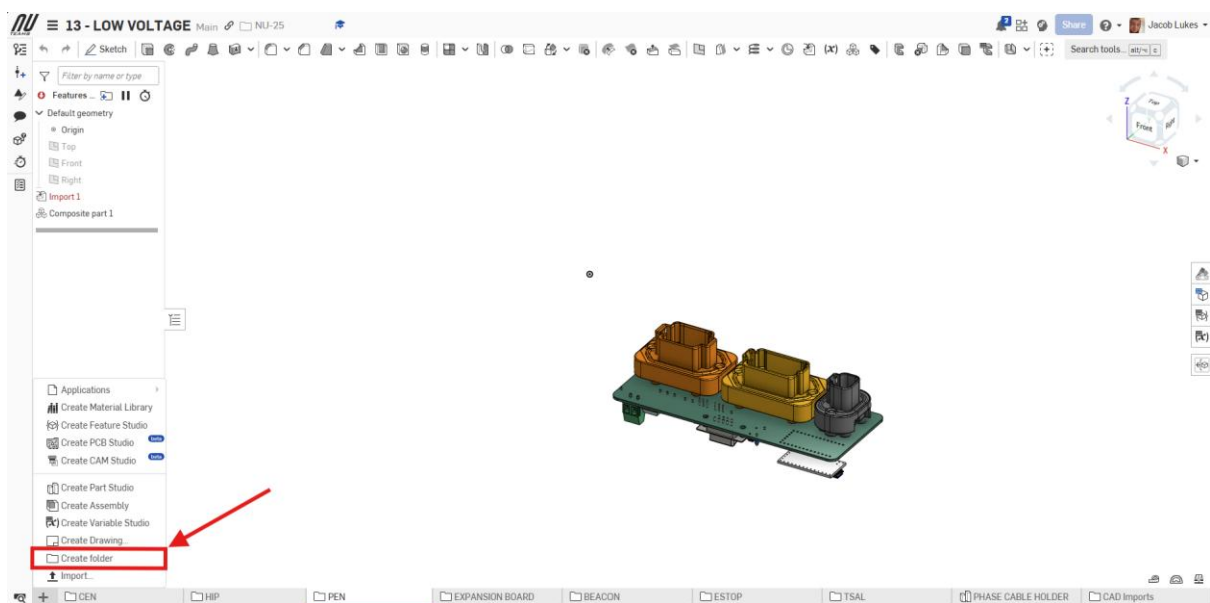
## How To Import PCB Into OnShape

- From within KiCad, export the PCB as a STEP file
- Open OnShape, ensure you are in the correct NU Racing Project Folder for the given year (eg: NU-25)
  - You may need to gain access from the Mechanical Department Leader first before being allowed to complete the following steps
    - If you are not allowed access, please get the enclosure designer to read this document
- Navigate to the desired document (eg: LOW VOLTAGE)

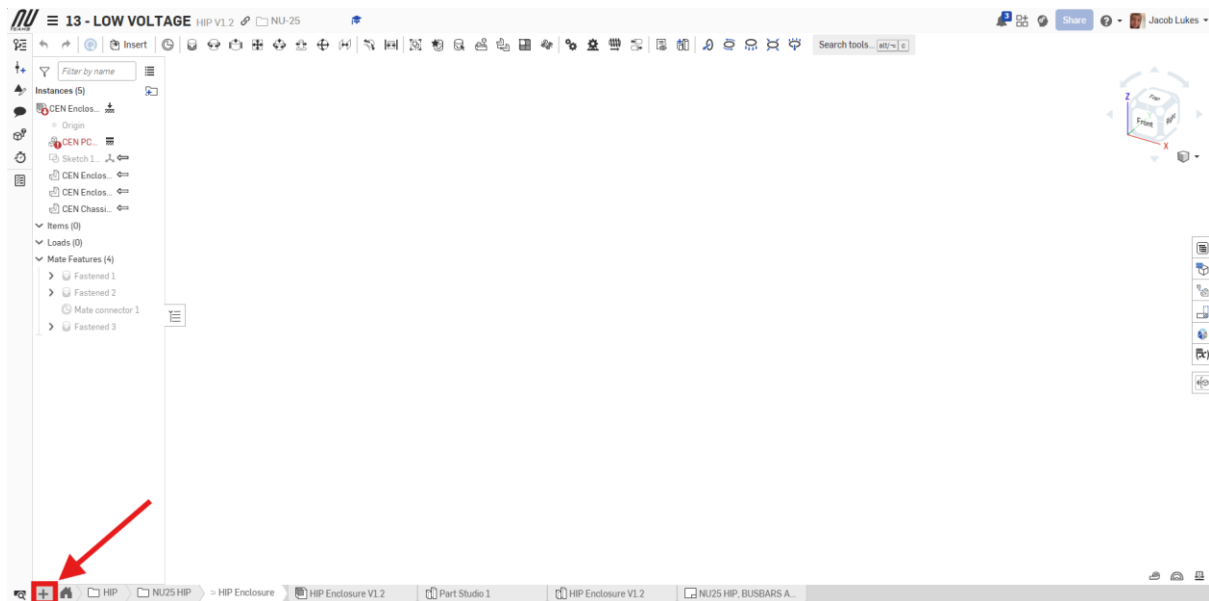
- Select the plus button on the bottom left-hand side of the screen



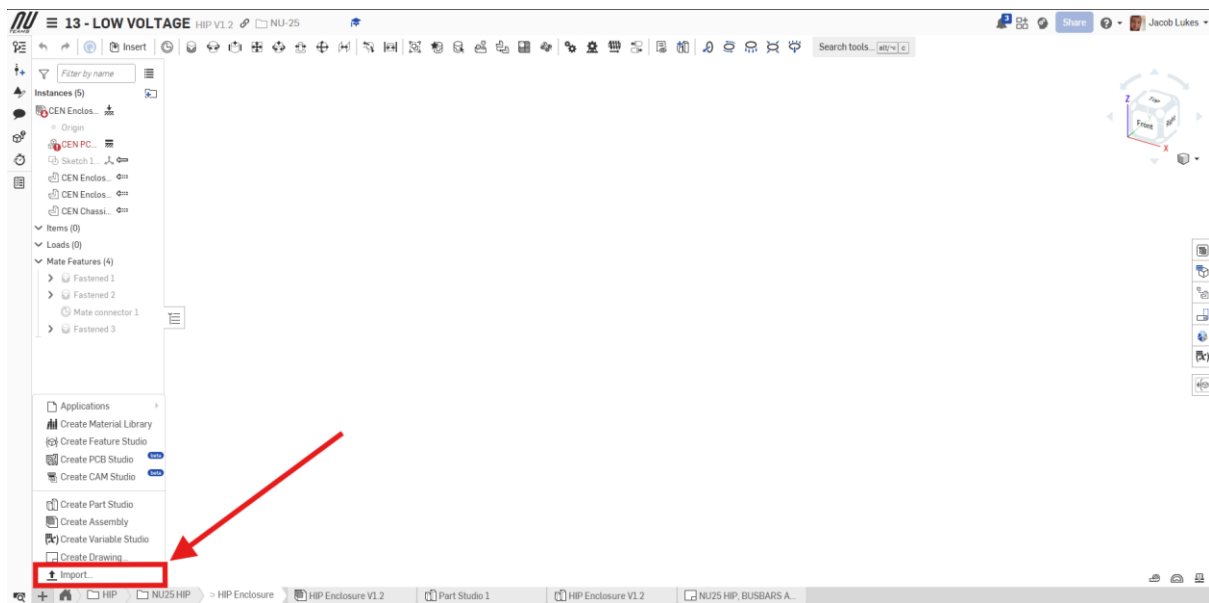
- Create a folder



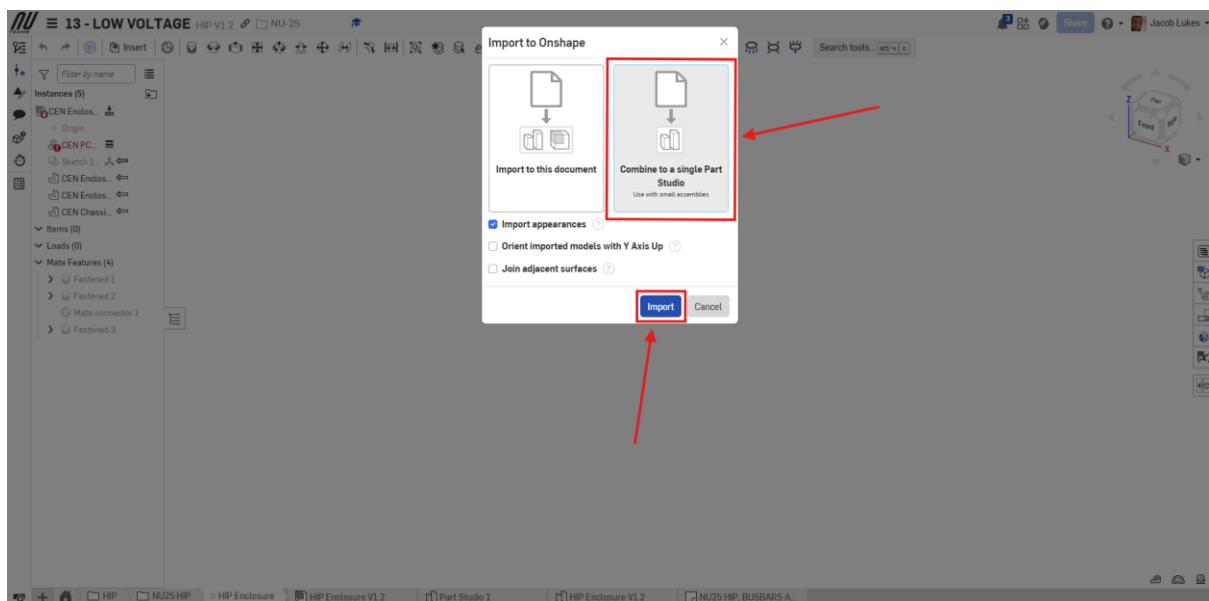
- Again select the plus button on the bottom left-hand side of the screen



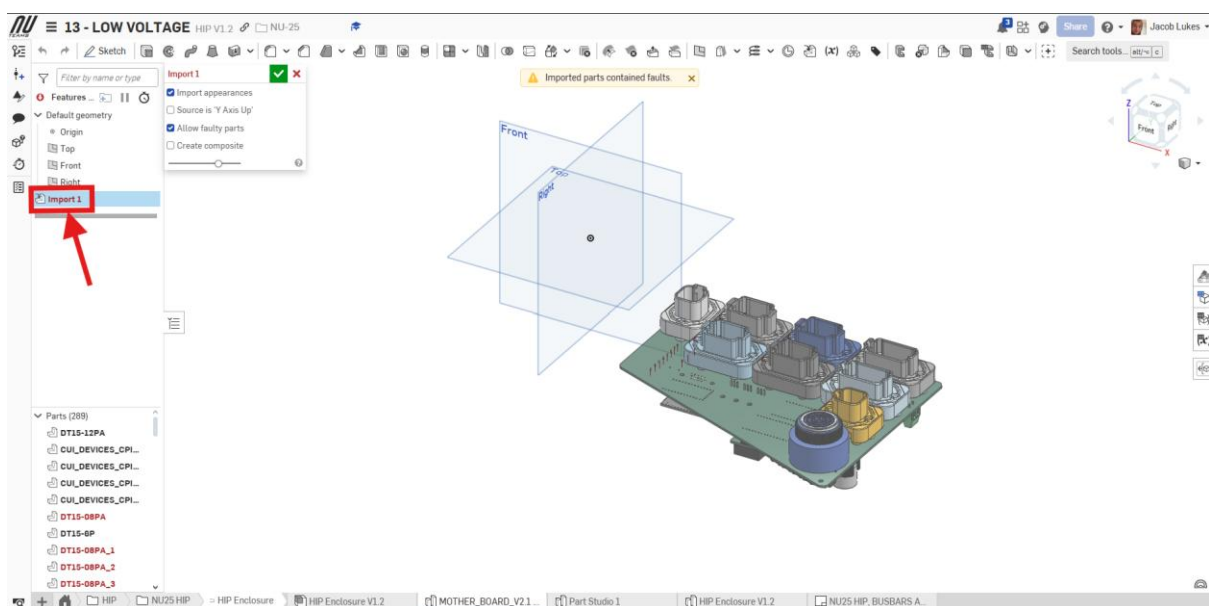
- Select 'Import'



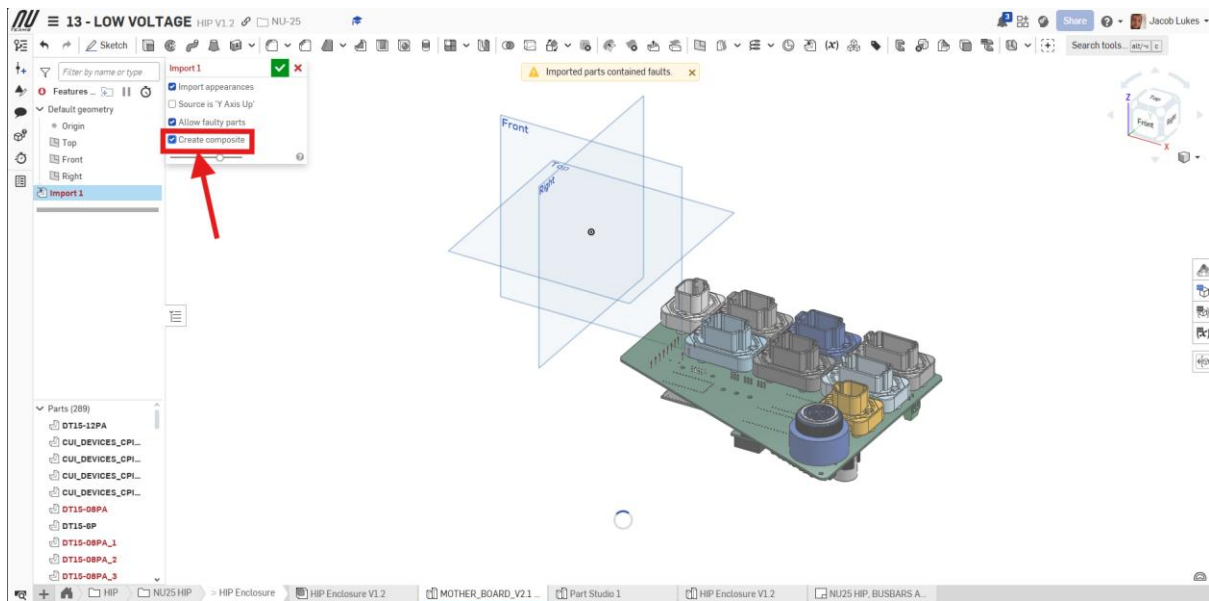
- Navigate to desired step file on PC
- Click '**Combine to a single Part Studio**'
  - **Tick:**
    - 'Import appearances'
  - **Untick:**
    - 'Orient imported models with Y Axis Up'
    - 'Join adjacent surfaces'
- Click 'import'



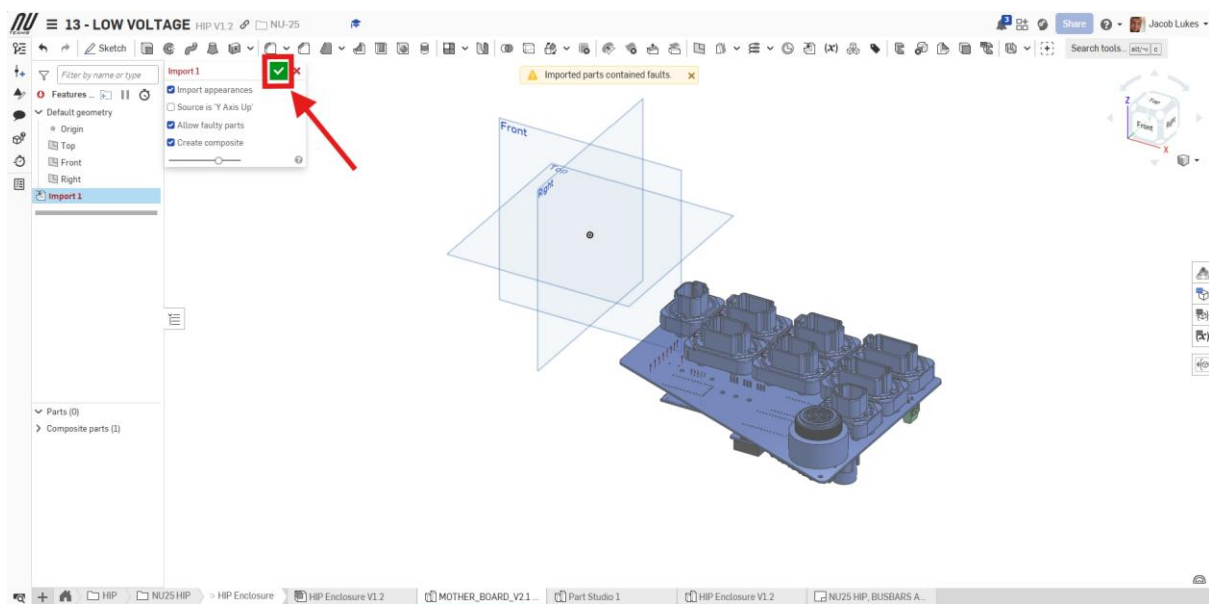
- **Double click the imported part**, usually 'Import 1's



- Tick '**Create composite**'

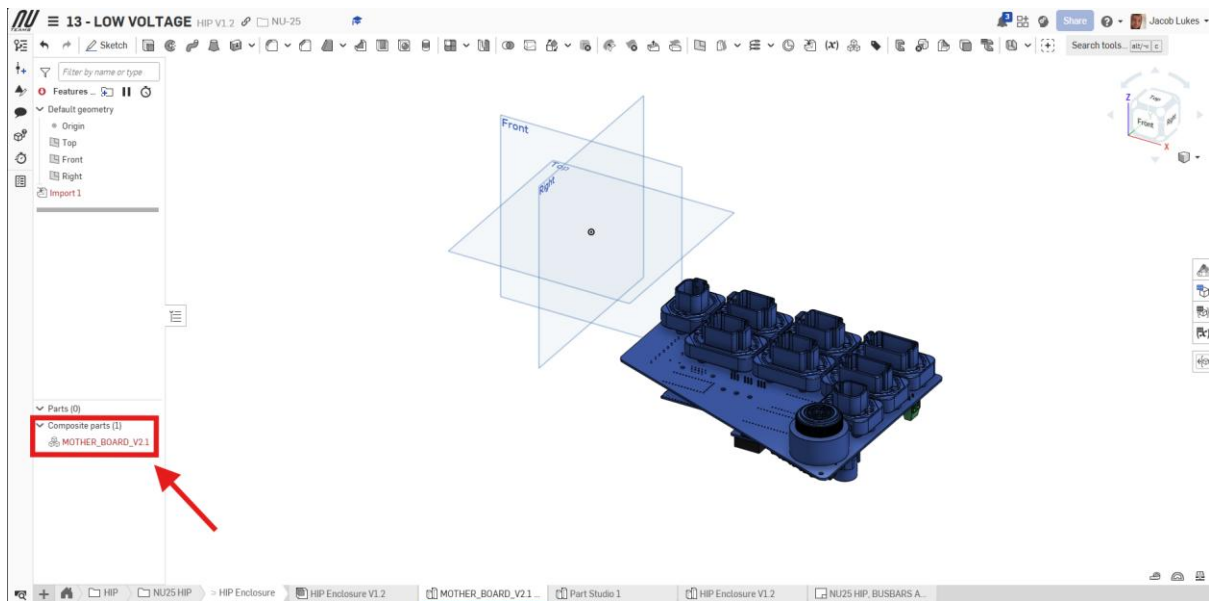


- Click the green check box



- Wait for it to think

- Ensure the composite part is the name of the given PCB

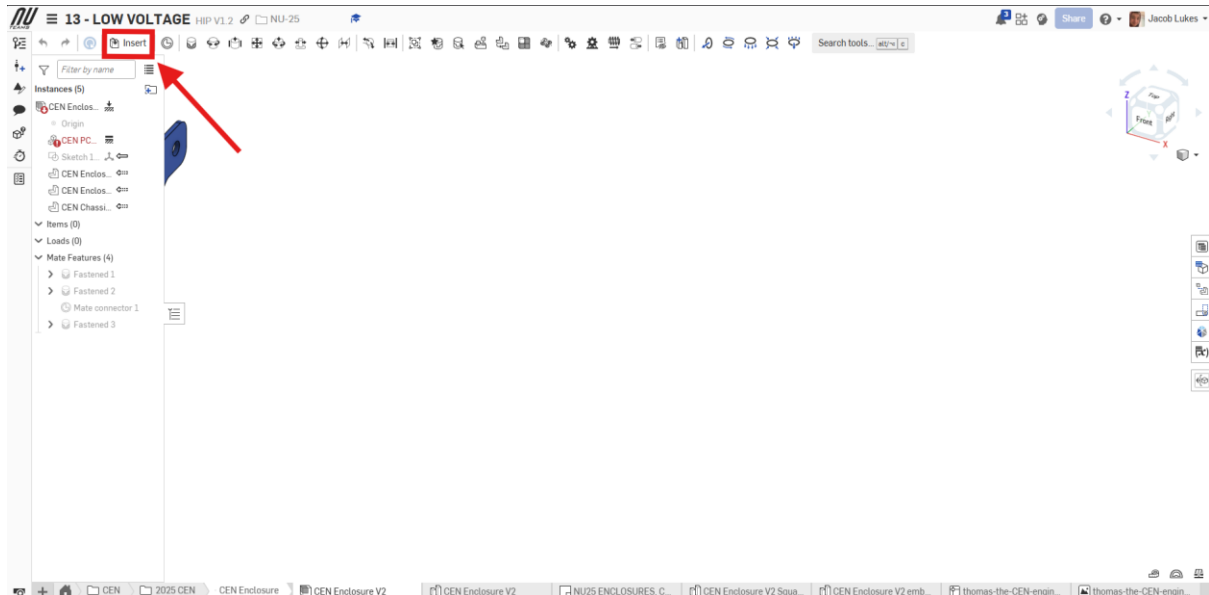


- Done!

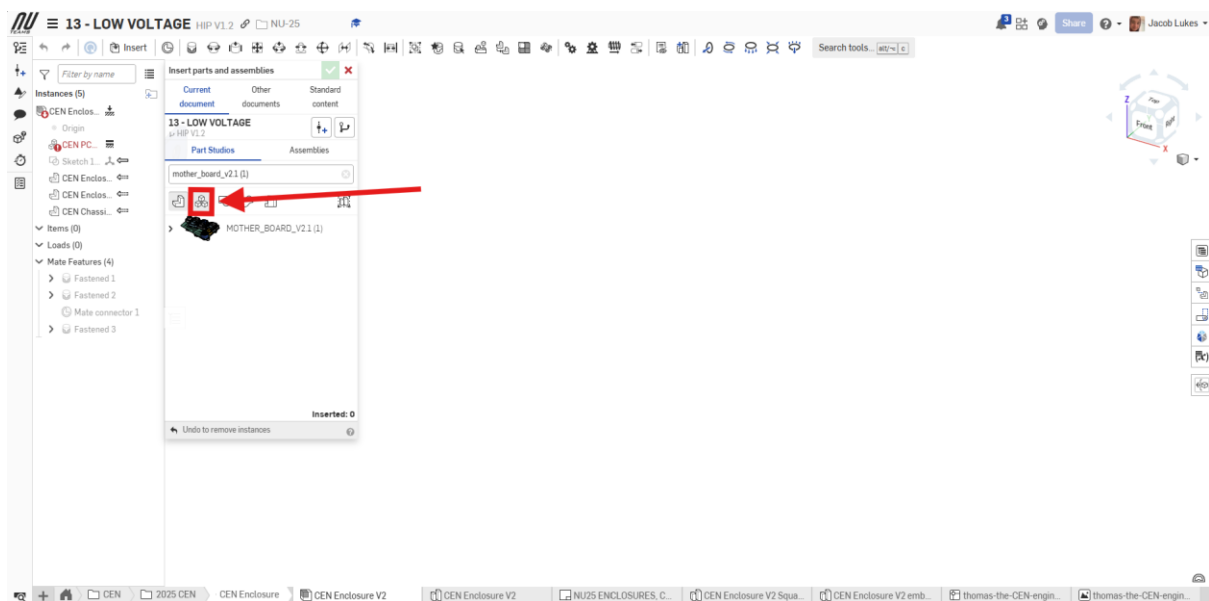


## How To Import PCBs Into an Assembly

- **Only complete this section if requested by a member from the mechanical team**
- Navigate to the desired Assembly
- Click 'Insert'



- Ensure to click 'Composite parts'



- Search for the desired PCB
- Clicking on the part will bring it into the Assembly